Huawei Access Node Optical Transport Failure SOP (JAY-ACC-35)

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_5519

Description

This Standard Operating Procedure (SOP) addresses the detection and resolution of Optical Transport Failures on Huawei Access Node JAY-ACC-35 located at the Jayanagar Metro Site, where the Role is Access and the CLI dialect is Huawei. The failure domain involves SFP/XFP degradation, fiber cuts, Rx/Tx power loss, which are indicative of ALARM_5519 severity. The underlying protocol and hardware mechanisms responsible for this failure domain are rooted in the Optical Transport Network's (OTN) Wavelength Division Multiplexing (WDM) technology, which transmits multiple signals over a single fiber optic cable. In the event of a failure, the OTN's supervisory channels may not function correctly, leading to a degradation of signal quality, which in turn triggers the ALARM_5519 alarm on the Huawei Access Node. This SOP is designed to isolate the root cause of the failure and apply corrective measures to restore the network's optical transport functionality.

Pre-requisite Checks

- Ensure that all necessary personnel are dispatched to the Jayanagar Metro Site to assist with the failure resolution.

- Verify that the redundant link status is operational before proceeding with the diagnostic and resolution steps.

- Confirm that the Huawei Access Node JAY-ACC-35 is accessible via the console or network interface.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: display optical interface, Expected Output: Optical interface information, including SFP/XFP type, wavelength, and signal quality, Description: This command retrieves information about the optical interface, including the type of SFP/XFP used, the wavelength, and the signal quality. The output of this command helps to isolate the root cause of the failure by indicating the presence of SFP/XFP degradation or other optical interface issues.

Command: display optical alarm, Expected Output: Alarm information, including ALARM_5519 and other optical-related alarms, Description:
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This command displays alarm information, including the ALARM_5519 alarm and other optical-related alarms. The output of this command helps to verify the presence of the ALARM_5519 alarm and determine the severity of the failure.

Command: display optical wavelength, Expected Output: Wavelength information, including the wavelength and channel power, Description: This command displays wavelength information, including the wavelength and channel power. The output of this command helps to verify the presence of fiber cuts or other optical-related issues.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical module, Description: Replace the SFP/XFP module on the Huawei Access Node JAY-ACC-35 with a new one. Ensure that the replacement module is compatible with the existing wavelength and optical interface configuration.

Step 2: Step: Check and repair fiber cuts, Command: check fiber status, Description: Verify the presence of fiber cuts and perform necessary repairs to restore the optical transport functionality. Ensure that the fiber cuts are properly spliced and the optical interface configured correctly.

Step 3: Step: Adjust Rx/Tx power settings, Command: modify optical power, Description: Adjust the Rx/Tx power settings to restore optimal signal quality. Ensure that the power settings are compatible with the existing optical interface configuration and wavelength.

Post-Check Verification

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Command: display optical interface, Expected Output: Optical interface information, including SFP/XFP type, wavelength, and signal quality, Description: Verify that the optical interface is functioning correctly and the signal quality has been restored.

Command: display optical alarm, Expected Output: No alarm information or the ALARM_5519 alarm cleared, Description: Verify that the ALARM_5519 alarm has been cleared or no alarm information is present.

Command: display optical wavelength, Expected Output: Wavelength information, including the wavelength and channel power, Description: Verify that the wavelength information is accurate and the channel power is within optimal range.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore optical module, Description: Restore the original SFP/XFP module to its previous configuration. Ensure that the replacement module is removed and the original one is properly secured in place.

- Step: Re-check fiber cuts, Command: check fiber status, Description: Verify the presence of fiber cuts and re-check the repairs performed. Ensure that the fiber cuts are properly spliced and the optical interface configured correctly.

- Step: Re-adjust Rx/Tx power settings, Command: modify optical power, Description: Re-adjust the Rx/Tx power settings to their original values. Ensure that the power settings are compatible with the existing optical interface configuration and wavelength.